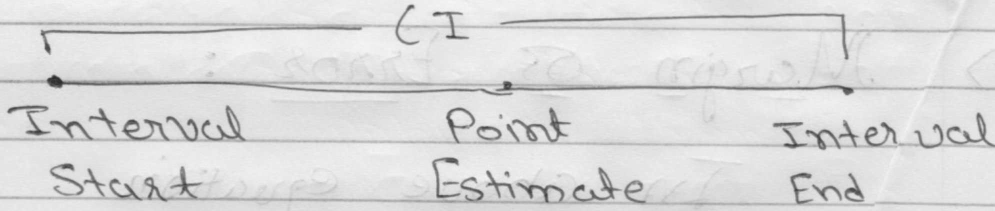
Confidence Intervals

⇒ **Confidence Intervals:**

A confidence Interval is an Interval within which we are Confident, with a certain percentage of confidence, the Population parameter will fall.



↳ $1-\alpha$ is the Level OF confidence.

↳ We are $(1-\alpha)\cdot 100\%$ confident that the population parameter will fall in Specified Interval.

↳ Most common values of α : 0.05, 0.01, 0.1

Confidence Interval =

$$\left[\text{Point Estimate} - \text{Reliability Factor} \cdot \text{Standard Error} , \right.$$

$$\left. \text{Point Estimate} + \text{Reliability Factor} \cdot \text{Standard Error} \right]$$

⇒ Margin Of Error :

In above equation

$\left(\text{Reliability Factor} \cdot \text{standard Error} \right)$ part is

called Margin Of Error Of ME.

$$\therefore ME = \text{Reliability Factor} \cdot \frac{\text{Standard Deviaton}}{\sqrt{\text{Sample size}}}$$

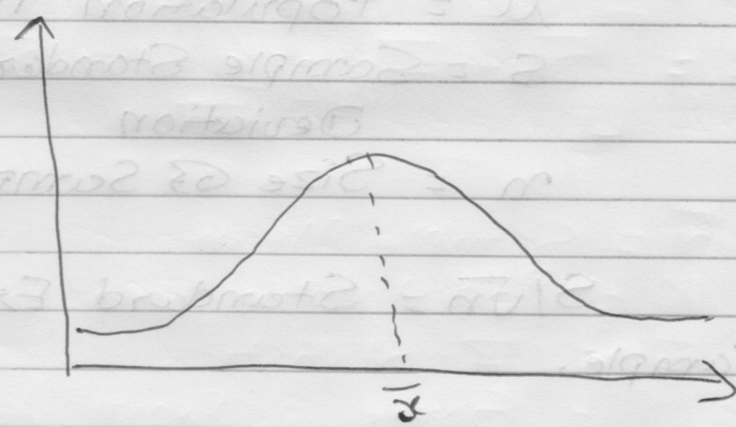
~~=> Situations and Confidence Interval Formulas according to Situation :~~

~~<1> When Population Variance is Know~~

=> Student's - T Distribution :->

The Student's - T Distribution is used predominantly for Creating Confidence Intervals and Testing Hypothesis with Normally Distributed Populations when the Sample Sizes are small.

The Student' - T Distribution has fatter tails and a lower Peak than Normal Distribution.



↳ It allows Inference through Small Samples with Unknown Population Variance.

↳ It has Huge Real life applications.

In the Same way that the Z-Statistics is related to the Standard Normal Distribution, the T-Statistics is related to the Student's - T Distribution.

$$t_{n-1,\alpha} = \frac{\bar{x} - \mu}{S/\sqrt{n}}$$

$n-1$ = degrees of freedom
α = Significance Level
$\bar{x}$ = Sample mean
μ = Population mean
S = Sample Standard Deviation
n = Size of Sample

$S/\sqrt{n}$ = Standard Error of Sample.

Usually for a Sample size of 'n', we have 'n-1' degrees of Freedom.

i.e. For a Sample of 20 Observations, Degrees of Freedom are 19.

<u>Note</u>: After 30 degrees of Freedom, the T-Statistics values become almost the Same as Z-Statistics.

"As a Rule of thumb, For a sample of size more than 50, we use Z-table insted of T-table."

⇒ Different Situations and How to calculate Confidence Interval for given Situation :

There can be two main situations when we calculate Confidence Intervals for a Population:

<1> Population Variance is known.

<2> Population Variance is Unknown.

⊙ Confidence Interval for a Population Mean with a known Variance :

$$CI = \left[\bar{x} - Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \ , \ \bar{x} + Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right]$$

$\bar{x}$ = Sample Mean
σ = Population Standard Deviation
Z = Z-Statistics
α = Same α from Level of Confidence $1-\alpha$.

∴ $Z_{\alpha/2}$ = Reliability Factor

$\frac{\sigma}{\sqrt{n}}$ = Standard error

$Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{2}}$ = Margin of Error

e.g.

Sample Mean $\bar{x}$ = 1,00,200
Population Standard Deviation (σ) = 15,000
Standard Error = 2,739
Sample Size = 30

$$\therefore \text{Standard Error} = \frac{\sigma}{\sqrt{n}} = \frac{15{,}000 \; \cancel{2739}}{\sqrt{30}} = 2739$$

Chosing Confidence Level of 95%

$\cancel{\alpha = 1}$ $\therefore 0.95 = 1 - \alpha$

$\therefore \alpha = 0.05$

Now, We Will find out $Z_{\alpha/2}$ From Standard Normal Distribution Table.

This table is available readily for use. (Search online for Standard Normal Distribution Table).

$$Z_{\alpha/2} = Z_{0.05/2} = Z_{0.025}$$

$\therefore$ Look for $1 - 0.025 = 0.975$ in table

Take sum of Column & Row Labels of 0.975 value in Table.

~~For 0.975 in table:~~
~~Column Label = 1.9~~

For 0.975 in table:
Row Label = 1.9
Column Label = 0.06

$$\therefore Z_{0.025} = 1.96$$

$$\therefore CI = [100200 - (1.96)(2739),$$
$$100200 + (1.96)(2739)]$$

$$\therefore CI = [94833, 105568]$$

Chosing Confidence Level of 99%:

$$\therefore 1-\alpha = 0.99$$
$$\therefore \alpha = 0.01$$

$$\therefore Z_{\alpha/2} = Z_{0.01/2} = Z_{0.005}$$

$\therefore$ Look for $1 - 0.005 = 0.995$ in table

For 0.995 in table:
Row = 2.5
Column = 0.08

$$\therefore Z_{0.005} = 2.58$$

$$\therefore CI = \left[100200 - (2.58)(2739),\ 100200 + (2.58)(2739)\right]$$

$$\therefore CI = [93135, 107206]$$

⊙ Confidence Interval For a Population Mean with an Unknown Variance:

$$CI = \left[\bar{x} - t_{n-1,\alpha/2}\frac{S}{\sqrt{n}},\ \bar{x} + t_{n-1,\alpha/2}\frac{S}{\sqrt{n}}\right]$$

$\bar{x}$ = Sample Mean
S = Sample Standard Deviation
t = T- statistics
α = Same α From Level of Confidence $1-\alpha$.
n = Sample Size
~~n -~~
~~t_{n-1}~~
$n-1$ = degrees of freedom.

$\therefore t_{n-1, \alpha/2}$ = Reliability Factor

$\frac{S}{\sqrt{n}}$ = Standard Error

$\therefore t_{n-1, \alpha/2} \cdot \frac{S}{\sqrt{n}}$ = Margin of Error

e.g. Sample Mean = 92,533
Sample Standard Deviation = 13932

Sample size = 9

$\therefore$ Standard Error = $\frac{S}{\sqrt{n}} = \frac{13932}{\sqrt{9}}$

$\therefore = 4644$

Chosing Confidence Level of 95%

$\therefore 1 - \alpha = 0.95$

$\therefore \alpha = 0.05$

Degrees of freedom = $n - 1 = 8$

$\therefore t_{n-1, \alpha/2} = t_{8, 0.025}$

Now, we will find out $t_{8, 0.025}$ From T-statistics Table.

This table is available readily for use.

In this table, Columns $= \alpha/2$
Rows $=$ Degrees of freedom.

$\therefore$ Look for a value associated with Row 8 & Column 0.025 in table.

$$\therefore \cancel{t_{n-1}} \; t_{8, 0.025} = 2.306$$

$$\therefore CI = \left[92{,}533 - (2.306)(4644), \; 92{,}533 + (2.306)(4644) \right]$$

$$\therefore CI = [81806, 103261]$$

⇒ Calculating Confidence Intervals For two Populations :

↳ In Some Cases, Samples we take from two populations will be dependent on each other.

↳ In other cases, Samples we take from two Populations will be Independent from each other.

e.g. Dependent Samples : When we are Researching Same Subject over time.
i.e. weight loss,
Blood samples.
Here, we are looking at the same person Before & After.

e.g. When working with ~~depe~~ Independent Samples, we have 3 cases:

<1> Population Variance is known.
<2> Population Variance is Unknown but assumed to be equal.

<3> Population Variance is Unknown but assumed to be different.

<1> Dependent Samples:

⊙ Confidence Interval for difference of two Means, Dependent Samples:

$$CI = \left[\bar{d} - t_{n-1,\alpha/2} \cdot \frac{S_d}{\sqrt{n}},\ \bar{d} + t_{n-1,\alpha/2} \cdot \frac{S_d}{\sqrt{n}}\right]$$

~~$\bar{d}$ = difference of Mean of Samples.~~
~~S_d = Standard Deviation of difference~~

$\bar{d}$ = Mean of difference of two Samples
S_d = Standard Deviation of difference of two Samples.
n = Sample size.

e.g. Checking effect of Magnesium tablet:

Patient	Before	After	Difference
1	2	1.7	-0.3
2	1.4	1.7	0.3
3	1.3	1.8	0.5
4	1.1	1.3	0.2
5	1.8	1.7	-0.1
6	1.6	1.5	-0.1
7	1.5	1.6	0.1
8	0.7	1.7	1
9	0.9	1.7	0.8
10	1.5	2.4	0.9

$\therefore \bar{d} = 0.33$

$S_d = 0.45$

Chosing Confidence Level = 95 %.

$\therefore \alpha = 0.05$

$n-1 = 9$

$\therefore t_{n-1, \alpha/2} = t_{9, 0.025}$

$= 2.26$

$$\therefore CI = \left[0.33 - (2.26)\frac{0.45}{\sqrt{10}}, \right.$$

$$\left. 0.33 + (2.26)\frac{0.45}{\sqrt{10}} \right]$$

$$\therefore CI = [0.07, 0.65]$$

<2> Indepent Samples :

- Confidence Interval for difference of Two means, Independent Samples, known Population Variance :

$$CI = \left[(\bar{x} - \bar{y}) - Z_{\alpha/2}\sqrt{\frac{\sigma^2_x}{n_x} + \frac{\sigma^2_y}{n_y}}, \right.$$

$$\left. (\bar{x} - \bar{y}) + Z_{\alpha/2}\sqrt{\frac{\sigma^2_x}{n_x} , \frac{\sigma^2_y}{n_y}} \right]$$

$\bar{x}, \bar{y}$ = mean of Samples

$$\sqrt{\frac{\sigma_x^2}{n_x} + \frac{\sigma_y^2}{n_y}} = \text{Variance of the Difference.}$$

e.g.

	Engineering	Management	Diff.
Size	100	70	-
Sample mean	58	65	(-7)
Sample Std.	10	5	1.16

$$\sigma_{diff}^2 = \frac{\sigma_x^2}{n_x} + \frac{\sigma_y^2}{n_y}$$

$$= \frac{100}{100} + \frac{25}{70}$$

$$= 1.36$$

$$\therefore \sigma_{diff} = \sqrt{1.36} = 1.16$$

$\therefore$ ~~CI = [(-7)~~

Chosing confidence Level = 95 %

$\therefore \alpha = 0.05$

$\therefore Z_{\alpha/2} = \cancel{Z_{0.05}} = Z_{0.025}$
$= 1.96$

$\therefore CI = \left[(-7) - (1.96)(1.16),\ (-7) + (1.96)(1.16) \right]$

$\therefore CI = [-9.28\ ,\ -4.72]$

- Confidence Interval for difference of two means, Independent Samples, Population Variance Unknown but Assummed to be equal :

$$CI = \left[(\bar{x} - \bar{y}) - t_{n_x+n_y-2,\, \alpha/2} \sqrt{\frac{S_p^2}{n_x} + \frac{S_p^2}{n_y}},\right.$$

$$\left. (\bar{x} - \bar{y}) + t_{n_x+n_y-2,\, \alpha/2} \sqrt{\frac{S_p^2}{n_x} + \frac{S_p^2}{n_y}} \right]$$

~~$\bar{x}, \bar{y}$ = Difference Of Mean~~

$\bar{x}, \bar{y}$ = Means Of Samples

S_p^2 = Pooled Variance

$$S_p^2 = \frac{(n_x - 1) S_x^2 + (n_y - 1) \cdot S_y^2}{n_x + n_y - 2}$$

~~e.g. Apples~~

e.g.

	A	B
Mean	3.94	3.25
Std.	0.18	0.27
Sample size	10	8

$$\therefore S_p^2 = \frac{(10-1)(0.18)^2 + (8-1)(0.27)^2}{10+8-2}$$

$$\therefore S_p^2 = 0.05$$

$\therefore$ Pooled Variance $S_p^2 = 0.05$

Pooled Standard Deviation $= 0.22$

Chosing confidence level $= 95\%$

$\therefore \alpha = 0.05$

$\therefore t_{n_x+n_y-2,\, \alpha/2} = t_{10+8-2,\, 0.05/2}$

$= t_{16,\, 0.025}$

$t_{16,\, 0.025} = 2.12$

$$\therefore CI = \left[(3.95 - 3.24) - (2.12) \cdot \sqrt{\frac{0.05}{10} + \frac{0.05}{8}},\ (3.95 - 3.24) + (2.12) \cdot \sqrt{\frac{0.05}{10} + \frac{0.05}{8}} \right]$$

$\therefore CI = [0.47, 0.92]$

⊙ Confidence Interval for difference of Two means, Independent Samples, Population Variance Unknown and Assumed to be Different :

$$CI = \left[(\bar{x} - \bar{y}) - t_{v, \alpha/2} \sqrt{\frac{S_x^2}{n_x} + \frac{S_y^2}{n_y}}, \right.$$

$$\left. (\bar{x} - \bar{y}) + t_{v, \alpha/2} \sqrt{\frac{S_x^2}{n_x} + \frac{S_y^2}{n_y}} \right]$$

$\bar{x}, \bar{y}$ = Mean of Samples
S_x^2, S_y^2 = Sample Variances

v = degrees of Freedom

$$v = \frac{\left(\frac{S_x^2}{n_x} + \frac{S_y^2}{n_y} \right)^2}{\left(\frac{S_x^2}{n_x} \right)^2 / (n_x - 1) + \left(\frac{S_y^2}{n_y} \right)^2 / (n_y - 1)}$$